

JEE MAIN REVISION MASTER: COMPLEX NUMBERS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Modulus, Argument & Polar / Euler Forms	7 PYQs	Complex Numbers
Standard	Cube Roots of Unity & Complex Equations	2 PYQs	Roots of Unity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Modulus, Argument & Polar / Euler Forms	$ z_1 + z_2 \leq z_1 + z_2 $ (Triangle Inequality), $z = r e^{i\theta}$
Cube Roots of Unity & Complex Equations	$1 + \omega + \omega^2 = 0$, $\omega^3 = 1$, $\omega^{3k+1} = \omega$, $\omega^{3k+2} = \omega^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Modulus, Argument & Polar / Euler Forms (Moderate Yield — 7 PYQs)

Core Principle: Modulus $|z| = \sqrt{a^2 + b^2}$. Principal argument $\theta = \text{Arg}(z) \in (-\pi, \pi]$. Properties: $|z_1 z_2| = |z_1| |z_2|$, $\arg(z_1 z_2) = \arg z_1 + \arg z_2$, $z \bar{z} = |z|^2$. Square root of $z = a+ib$: $\pm \sqrt{\frac{|z|+a}{2}} + i \text{sgn}(b) \sqrt{\frac{|z|-a}{2}}$.

■ **Caution / JEE Pitfall:** Principal argument is strictly defined in the interval $(-\pi, \pi]$; do not give values in $[0, 2\pi]$ if principal argument is requested.

■ **Speed Trick:** The triangle inequality $|z_1| - |z_2| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ yields the maximum and minimum distance on Argand diagrams instantly.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\sum_{n=1}^4 (-1)^{2n} \cdot i^{2n} = \dots$ " [Ans: C]
- [2024 Shift] Q#3 (MCQ): "The real value of ' α ' for which $(1 - i \sin \alpha) / (1 + 2i \sin \alpha)$ is purely real is..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "The modulus of the complex number $\frac{(1+i)^2(1+3i)(2-6i)(2-2i)}{(2-6i)(2-2i)}$ is..." [Ans: N/A]

3.2 Cube Roots of Unity & Complex Equations (Standard Yield — 2 PYQs)

Core Principle: Cube roots of unity are $1, \omega, \omega^2$ where $\omega = \frac{-1 + i\sqrt{3}}{2}$ and $\omega^2 = \frac{-1 - i\sqrt{3}}{2}$. Fundamental relations: $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$. Factors: $a^3 - b^3 = (a-b)(a - b\omega)(a - b\omega^2)$; $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$.

■ **Caution / JEE Pitfall:** When reducing higher powers of ω , divide the exponent by 3 and keep ONLY the remainder: $\omega^{100} = \omega^{3(33)+1} = \omega^1 = \omega$.

■ **Speed Trick:** The three cube roots of unity form the vertices of an equilateral triangle inscribed in a unit circle centered at the origin on the Argand plane.

Representative Source PYQs:

• **[2025 Shift] Q#2 (MCQ):** "If Z_1 and Z_2 are two non-zero complex numbers, then which of the following is not true?..."

[Ans: N/A]

• **[2022 Shift] Q#1 (MCQ):** "If $3x + i(4x - y) = 6 - i$ where x and y are real numbers, then the values of x and y are respectively,..."

[Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\sum_{n=1}^{\infty} (-1)^{4n} \cdot i^{2n} =$$

- (A) 2
- (B) -i
- (C) 0 [Correct]
- (D) i

Correct Answer: C

Solution Analysis: $(-1)^{4n}$ is always equal to 1, because the power $4n$ is even. So, $(-1)^{4n} \cdot i^{2n} = 1 \cdot i^{2n} = i^{2n}$. Now use $i^2 = -1$. $i^{2n} = (i^2)^n = (-1)^n$. So the given sum becomes $\sum_{n=1}^{\infty} (-1)^n = 1(-1)^n$. Now put $n=1, 2, 3, 4$: $(-1)^1 + (-1)^2 + (-1)^3 + (-1)^4 = -1 + 1 - 1 + 1 = 0$. So the correct option is 0 Option C.

Benchmark Problem 2: KCET 2022 (Q#1)

If $3x + i(4x - y) = 6 - i$ where x and y are real numbers, then the values of x and y are respectively,

- (A) 3, 9
- (B) 2, 4
- (C) 2, 9
- (D) 3, 4

Correct Answer: None

Solution Analysis: $3x + i(4x - y) = 6 - i$ On comparing real and imaginary parts of LHS and RHS, we get $3x = 6$ and $4x - y = -1 \Rightarrow 3x = 6 \Rightarrow x = 2$. Put $x = 2$ into $4x - y = -1$, we get $4 \times 2 - y = -1 \Rightarrow 8 - y = -1 \Rightarrow y = 9$.